

The Role of Synaptic Matrix Attributes in Spiking Patterns of Neural Networks

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1 Introduction

1.1 Motivation

In computational neuroscience, the structure of neural networks is an important area, since it explores how connectivities can influence the information processing of neurons. Understanding how neurons are connected, how they interact, and how these interactions can lead to diverse and complex behaviours is a motivation in ongoing research in this area. The patterns of connectivity among neurons, like their topological and geometric characteristics, play a crucial role in shaping the emergent dynamics observed in biological systems (Bullmore and Sporns [2009]).

Although much of traditional modelling focuses on stimulus-driven responses, many neural circuits exhibit rich spontaneous activity even in the absence of external input. Capturing such internally generated dynamics requires models that are both biologically realistic and computationally tractable (Vogels et al. [2005]).

The Izhikevich model offers an effective compromise between biological plausibility and efficiency. It reproduces a wide range of spiking patterns observed in real neurons while remaining suitable for large-scale network simulations (Izhikevich [2003]). In this study, we use this model to explore how different patterns of synaptic connectivity influence global network dynamics.

1.2 Biological Background

There are a few biological concepts that require a brief introduction.

- **Spiking:** Neurons communicate through discrete electrical events known as spikes, which are rapid rises and falls in membrane potential, where the membrane potential is the voltage difference between the inside and outside of a neuron, maintained by ion concentration gradients and selective membrane permeability. Depolarization occurs when this membrane potential becomes less negative, often because of positive ions entering the cell, and a spike is triggered when depolarization exceeds a threshold, allowing the neuron to transmit a signal. This spiking mechanism is a fundamental method by which information is encoded and transmitted in the brain. Both the rate and timing of spikes carry meaningful signals, allowing the brain to perform complex tasks (Dayan and Abbott [2001]).
- **Excitatory and inhibitory neurons:** The neurons in the brain can be partitioned into two functional groups, excitatory and inhibitory. Excitatory neurons increase the spiking probability of the postsynaptic neuron, typically through the release of glutamate. In contrast, inhibitory neurons reduce the likelihood of firing by releasing neurotransmitters such as GABA. The balance between excitation and inhibition is essential for stable and flexible brain activity (Isaacson and Scanziani [2011], Dayan and Abbott [2001]). There is a famous principle called Dale's law, which states that each neuron releases the same type of neurotransmitter in all its synapses, implying that the neuron is either inhibitory or excitatory (Catsigeras [2013]). In this study, three types of simulated cortical neurons are modelled to approximate the behaviour of actual biological cells: regular spiking neurons (RS), fast spiking neurons (FS), and low-threshold spiking neurons (LTS). Inspired by biological characteristics, these models are mathematical representations that replicate key features of their spiking dynamics.

- **Regular spiking neurons (RS)**: These neurons are modelled to mimic the behaviour of typical excitatory neurons in the cortex. When simulated with a depolarizing current above threshold, RS cells respond with an initial high-frequency spike output and rapidly decline to a much lower steady firing rate (Connors and Gutnick [1990]).
- **Fast spiking neurons (FS)**: Modelled after inhibitory neurons, these cells generate high-frequency firing without slowing down under conditions similar to those of RS cells, which is characteristic of certain inhibitory neurons in the cortex. When simulated near the threshold, FS cells typically exhibit a gradual depolarizing voltage ramp before firing spikes with long delays (Gibson et al. [1999]).
- **Low-threshold spiking neurons (LTS)**: These inhibitory neurons are modelled to show larger spikes, show an adaptation of the firing frequency, and never generate a long delay before spiking, similar to biological LTS neurons (Gibson et al. [1999]).
- **Network dynamics**: Neural networks in the brain are characterized by dynamic and recurrent interactions among large populations of excitatory and inhibitory neurons. These networks exhibit emergent activity patterns that are shaped by both external inputs and intrinsic connectivity (Vogels et al. [2005]). Rather than reflecting the behaviour of individual cells, network activity is shaped by collective dynamics, including balanced excitation and inhibition. One important mechanism underlying neural network dynamics is plasticity, which means that repeated, near-synchronous activation of both pre- and postsynaptic neurons produced an increase in the strength of the synaptic input of the stimulated connections (Magee and Grienberger [2020]).

1.3 Topological Considerations

This section introduces some topological concepts and principles in network connectivities which influence the dynamics and behaviour of the neural network.

- **Sparsely connectivity**: In real life, neurons in cortical networks are very sparsely connected, and sparse connectivity allows efficient training in networks with fixed excitatory and inhibitory nodes (Fruengel and Oberlaender [2025]).
- **Log-normal connection strength**: When applying statistical methods to a large visual cortex data set in developing rats, the probability density function of the synaptic connection strengths is best fitted by a log-normal function (Song et al. [2005]).
- **Small-world network**: Topologically, small-world property is a combination of high levels of local clustering among nodes of a network and short paths that globally link all nodes of the network. In other words, most nodes in the system maintain only a few direct connections, while all nodes are linked through relatively few intermediate steps. Network analyses of several species anatomical connection data have recognized small-world architecture. In addition, the functional connections between the cortical areas are also consistent with a small-world network (Bullmore and Sporns [2009]). Sometimes, to construct a small-world network, the Watts–Strogatz model is applied. The model is built on regular lattices and 'rewired' to insert increasing amounts of disorder (Watts and Strogatz [1998]).
- **Scale-free graph**: Mathematically, if a distribution follows a power law, the probability is given by $\text{prob}(k) \sim k^{-\lambda}$. In biological system, λ often ranges between 2 and 3, which leads to a very gradual decay of the distribution, and networks with such a 'heavy tail' are called 'scale-free' networks. Normally, 'scale-free' refers to the degrees of nodes (Bullmore and Sporns [2009]). Barabási-Albert model is utilized to simulate 'scale-free' features by continuously adding new vertices while attaching them preferentially to sites that are already well connected (Barabási and Albert [1999]).

1.4 Izhikevich Model

The Hodgkin-Huxley model is a biophysically detailed mathematical model that describes how spiking in neurons is initiated and propagated through voltage-gated ion channels (Hodgkin and Huxley [1952]). Due

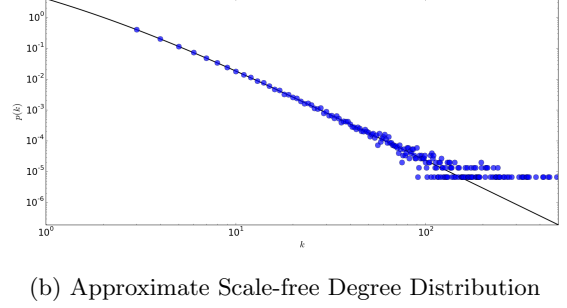
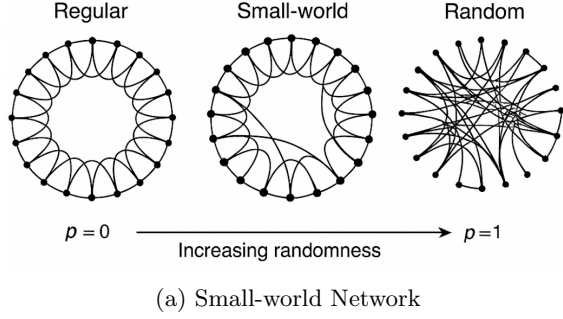


Figure 1: Illustration of two common network structures.

to the computational expense of Hodgkin-Huxley-type models, the Izhikevich model simplifies them into a two-dimensional system of ordinary differential equations; relying on just four parameters (a, b, c, d), it can reproduce the spiking and bursting behaviours of various cortical neuron types (Izhikevich [2003]). The ODEs are as follows:

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I, \quad (1)$$

$$\frac{du}{dt} = a(bv - u), \quad (2)$$

with the reset condition:

$$\text{if } v \geq 30 \text{ mV, then } \begin{cases} v \leftarrow c, \\ u \leftarrow u + d. \end{cases}$$

where v represents the membrane potential of neuron and u represents a membrane recovery variable.

In typical setups, neurons are divided into a biologically plausible 4:1 ratio of excitatory (RS) to inhibitory (FS) types. Each neuron receives both synaptic and noisy thalamic input. Synaptic interactions are captured in a weight matrix, with neurons coupled as pulse-coupled units where spikes instantaneously affect postsynaptic targets. The Izhikevich model uses a random uniform distribution, $\mathcal{U}(0, 0.5)$ for excitatory synapses and $\mathcal{U}(-1, 0)$ for inhibitory synapses, to generate connection strengths (Izhikevich [2003]).

1.5 Research Questions

Using this Izhikevich model, I want to investigate how network connectivity, synaptic properties, and cell type composition affect population-level spiking dynamics. So, this research includes four specific research questions:

- **How does network sparsity influence the frequency and distribution of spiking activity?**
In previous work, Kim and Lim stated that network rhythms only emerge when connectivity exceeds a critical threshold, implying that if the connection density falls below this value, sparse synchronization and rhythms collapse (Kim and Lim [2014]). In addition, Harris et al. demonstrated that decreasing sparsity reduces the magnitude of eigenspectral outliers, weakening the network's ability to sustain coherent rhythmic activity (Harris et al. [2023]). Therefore, I propose that as the synaptic matrix S become sparser, global rhythmic structure weakens.
- **What is the effect of scaling excitatory and inhibitory synaptic weights on the global rhythms and how sensitive is the network to changes in inhibitory/ excitatory strength?**
Sadeh and Clopath showed that changes in the synaptic strength of inhibitory connections alter not only the firing rates of participating neurons but also the formation and stability of spiking assemblies, which indicates non-linear effects (Sadeh and Clopath [2021]). And Rubin et al. claimed that the dynamics of neuronal networks are highly sensitive to the precise balance of excitatory and inhibitory synaptic strengths. Balanced synaptic input ensures both selectivity and robustness to noise, while deviations from this balance result in fragile network behaviour (Rubin et al. [2017]). Thus, I expect a change in the firing rate and sensitivity when modifying the synaptic weights strength.

- **How do variations in neuron type ratios (e.g. RS, FS, LTS) influence overall spiking behaviour?** In previous researches, Destexhe tested the Izhikevich model with RS, FS, and LTS neurons, and concluded that the presence of LTS cells increased the excitability of the network, making it possible to generate asynchronous irregular dynamics with a relatively modest network size (Destexhe [2009]). Vierling-Claassen et al. tested on a model that includes RS, FS, and LTS cells, and found that including LTS cells in the network reduced the probability of RS spike at higher drive frequencies, while adding LTS cells did not disrupt the fundamental gamma-band resonance driven by FS interneurons (Vierling-Claassen et al. [2010]). Tomov investigated networks composed of RS and other excitatory types plus FS or LTS inhibitory interneurons and discovered that networks with LTS more strongly supported self-sustained activity than FS, and that changing the ratios of excitatory types could alter both excitatory and inhibitory neuron firing rates depending on the inhibitory subtype (Tomov et al. [2014]). Hence, I expect a variation in the firing rate when LTS cells are introduced and when the inhibitory ratios are changed.
- **How do different synaptic weight distributions and network topologies influence emergent spiking dynamics?** Li et al. claimed that networks with a long-tailed distribution of the average outgoing synaptic weights exhibit bursting and modulation of firing rates in response to changes in inhibitory strength, while networks with a normal distribution do not burst and show little change in firing rate under the same conditions (Li et al. [2022]). In terms of topological structures, in Shanahan’s research, small-world networks tend to enhance the dynamical complexity (Shanahan). Kriener said that the average rate in all four networks is hardly affected by the underlying topology or weight distribution of the networks, while the variances of the population activity are very different (Kriener et al. [2009]). Based on these, I aim to compare spiking patterns across networks with different weight distributions and topologies, focusing on their influence on rhythm and variability.

2 Methods

The model is simulated for 10 iterations and each iteration runs for 1000 ms. During each iteration, the spiking activity of neurons is recorded based on the principles of the Izhikevich model.

2.1 Neuron Models

In this research, a total of 100 neurons are simulated, comprising three types of cortical cells. The parameters for these neuron types, according to the Izhikevich model, are set as follows:

Neuron Type	a	b	c	d
RS neurons	0.02	0.2	-65 mV	8
FS neurons	0.1	0.2	-65 mV	2
LTS neurons	0.02	0.25	-65 mV	2

Table 1: Izhikevich model parameters for different neuron types.

2.2 Network Structure

When initializing the synaptic weight matrix W , four different network structures are considered, and the corresponding Python code used to construct these networks is provided.

- **Random network (with desired density):** As in the original Izhikevich model, the synaptic weight matrix is initialized using a random uniform distribution. In my implementation, I introduce an additional parameter, the connection density, that controls the proportion of existing connections to achieve a desired level of sparsity in the network. Meanwhile, I extend the model to support multiple weight distributions, including uniform, normal, and log-normal. Among these, the log-normal distribution is considered particularly biologically realistic.

```

for i in range(N):
    for j in range(N):
        if i != j and np.random.rand() < desired_density:
            # the distribution selected
            if dist == 'uniform':
                W[i, j] = np.random.uniform(0, 1)
            elif dist == 'normal':
                W[i, j] = np.abs(np.random.normal(loc=0.5, scale=0.2))
            elif dist == 'lognormal':
                W[i, j] = np.random.lognormal(mean=0.0, sigma=1.0)

```

- **Lattice network:** The lattice preserves structural regularity. Each neuron is only connected to its neighbours. The variable radius controls how many neighbours on each side the neuron is connected to.

```

radius = 2
for i in range(N):
    for j in range(1, radius + 1):
        right = (i + j) % N
        left = (i - j) % N
        W[i, right] = np.random.rand() * 0.5
        W[i, left] = np.random.rand() * 0.5

```

- **Small-world network:** The modelling with this fundamental property of the brain organization makes the network biologically plausible. To help realise the small-world structure in Python code, Watts–Strogatz model is applied.

```

G = nx.watts_strogatz_graph(N, k=4, p=0.1)
W = nx.to_numpy_array(G) * np.random.rand(N, N) * 0.5

```

- **Randomly wired scale-free network:** ‘Scale-free’ normally refers to the degrees of nodes, but this paper explores ‘scale-free’ weights instead of degrees, since the research focuses on connection strength heterogeneity. Direct coding for Barabási-Albert model can only generate scale-free degrees, so the scale-free weights are transformed from degrees. Then the model masks a completely random connection pattern with the scale-free weights by shuffling two matrices element-wisely.

```

k_sf=int(np.ceil(desired_density*N))
G_sf = nx.barabasi_albert_graph(N, k_sf)
sf_weights = np.random.rand(G_sf.number_of_edges())*0.5

conn_mask = (np.random.rand(N, N) < 1.0)
np.fill_diagonal(conn_mask, 0)

conn_indices = np.transpose(np.nonzero(conn_mask))

num_weights = min(len(sf_weights), len(conn_indices))
np.random.shuffle(sf_weights)
np.random.shuffle(conn_indices)

rows, cols = conn_indices[:num_weights].T
W[rows, cols] = sf_weights[:num_weights]

```

After constructing the synaptic weight matrix W , the model enforces Dale’s law. Due to their nature, which decreases and increases the probability of firing, respectively, inhibitory and excitatory neurons have specific signs (Markram et al. [2004]).

```

for i in range(N):
    if types[i] in ['FS', 'LTS']:
        W[i, :] *= -1
    else:
        W[i, :] = np.clip(W[i, :], 0, None)

```

When working on different topologies, to make the result comparable across various networks, the model implements normalization. There are two normalizations:

- L1 normalization: L1 regularization tends to set many coefficients to exactly zero, causing sparse solutions (Tibshirani [1996]).

$$\mathbf{w}_{\text{normalized}} = \frac{\mathbf{w}}{\|\mathbf{w}\|_1} = \frac{\mathbf{w}}{\sum_{j=1}^n |w_j|}$$

- L2 normalization (Euclidean norm): Compared to L1, L2 regularization shrinks coefficients toward zero but rarely sets them exactly to zero.

$$\mathbf{w}_{\text{normalized}} = \frac{\mathbf{w}}{\|\mathbf{w}\|_2} = \frac{\mathbf{w}}{\sqrt{\sum_{j=1}^n w_j^2}}$$

2.3 Simulation Process

The simulation process is the primary part of the Izhikevich model, and the following is the Python code translated from the Matlab code in Izhikevich [2003]. The variable, noise standard deviation, is set to add noise to the input current.

```

for i in range(N):
# get_izh_params defined for neuron type in Section 2.1
    a[i], b[i], c[i], d[i] = get_izh_params(types[i])
    u[i] = b[i] * v[i]

# Realize Izhikevich ODEs
for t in range(T):
    I = np.dot(W, spikes[:, t-1]) + np.random.normal(10, noise_std, N)
    fired = v >= 30
    spikes[fired, t] = 1
    v[fired] = c[fired]
    u[fired] += d[fired]

    v += dt * (0.04 * v**2 + 5 * v + 140 - u + I)
    u += dt * a * (b * v - u)

```

To mimic the cerebral network more precisely, the model considers plasticity. One of the learning rules that capture this phenomenon is Hebbian learning, and it is calculated based on the principle that 'neurons that fire together, wire together' (Siri et al. [2007]).

```

for t in range(1, T):
    pre = spikes[:, t-1]
    post = spikes[:, t]
    # lr defined as the step size for updating synaptic weights
    delta_w = lr * np.outer(post, pre)
    W += delta_w

```

If applying Hebbian learning, the outcome W is required to apply Dale's rule again to ensure that each neuron maintains a consistent type of function.

2.4 Measures

The spiking activity of the simulated neurons is analysed using several measures, combining both graph-based representation and data analysis techniques.

- Graph structure analysis: There are four attributes that come directly from the weight matrix generated:
 - Average degree: Calculate the average strength of connections per node.
 - Clustering: Calculate the average clustering coefficient. The clustering coefficient quantifies the number of connections that exist between the nearest neighbours of a node as a proportion of the maximum number of possible connections. In other words, the clustering coefficient measures the tendency of the nodes to form triangles. Higher values mean more local connectivity (Bullmore and Sporns [2009]).

$$C_i = \frac{1}{k_i(k_i - 1)} \sum_{j,k} (w_{ij} \cdot w_{jk} \cdot w_{ki})^{1/3}$$

- Strength mean: Calculate the average outgoing synaptic strength across neurons. For example, if $W[i,j]$ denotes the strength of the synaptic connection from neuron i to neuron j , then outgoing connections refers to all synapses neuron i sends to others.
- Strength standard error of the mean: This quantifies the uncertainty or variability in the estimate of the mean synaptic strength.

$$\text{SEM} = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}}{\sqrt{n}}$$

- Spontaneous spiking activity spectrum: It plots the spiking time for all neurons over a 1000 ms interval.
- Firing rate analysis: These metrics are derived from the spiking activity of neurons.
 - Average rate: Calculate the mean firing rate across the entire population.
 - Coefficient of variation (CV): Measures variability in firing rate across neurons. A higher CV indicates more heterogeneous activity, while a lower CV suggests more uniform firing.

$$\text{CV} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}}{\mu}$$

- Mean firing rate distribution spectrum: After computing the average firing rate for each neuron, this spectrum illustrates the distribution of these mean values.
- Fourier power spectrum: The Fast Fourier Transform is a computational algorithm used to convert a signal from the time domain into the frequency domain. It decomposes a complex signal into a combination of sine and cosine waves of varying frequencies. The Discrete Fourier Transform formula is shown below:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-2\pi i \frac{kn}{N}}, \quad k = 0, 1, \dots, N-1$$

where

- $x[n]$: the signal in the time domain at sample index n
- $X[k]$: the frequency component at index k
- N : total number of time points
- $e^{-2\pi i \frac{kn}{N}}$: complex exponential basis function

I compute the Fourier power spectrum to quantify the signal’s power at each frequency by taking the squared magnitude of the Fourier transform coefficients. This analysis enables the identification of dominant frequencies in the neural activity and facilitates comparisons of rhythmic patterns across different neuron types and network configurations. Below is the code used to apply Fast Fourier Transforms:

```
# Average population activity over multiple simulations
# neuron_indices is the group of indices of a specific neuron type
# all_spike_matrices includes times of spiking for all neurons.
summed_signal = sum(
    spikes[neuron_indices].sum(axis=0) for spikes in all_spike_matrices
) / len(all_spike_matrices)

# Perform Fourier Transform
T = len(summed_signal)
freqs = fftfreq(T, dt / 1000) # frequency in Hz and dt=1.0
power = np.abs(fft(summed_signal))**2

# Keep only positive frequencies
freqs_pos = freqs[:T // 2]
power_pos = power[:T // 2]
power_pos[0] = 0 # Remove DC component, i.e. the zero-frequency component
```

3 Results

With respect to different attributes of the synaptic weight matrix, the following sections present the results of experiments.

3.1 Effect of Connection Density

3.1.1 Settings

To investigate how sparsity affects the frequency or distribution of the spike, I conducted seven experiments using the model outlined in Section 2. The neurons in these experiments adhered to the default settings of the original Izhikevich model. The synaptic weight matrix was generated from a random network following a normal distribution, with the sparsity of connections varied by setting the desired density to {0.05, 0.1, 0.2, 0.4, 0.5, 0.6, 0.8}. No post-generation normalisation was applied. The model was then operated following the rules of the Izhikevich model, with plasticity incorporated.

3.1.2 Result and Analysis

In this experiment, I mainly collected the overall graph structure data (average degree, clustering, strength mean and strength standard error of the mean) and the firing rate data (average rate and CV). Additionally, separate data for RS and FS neurons were recorded. The specific metrics collected are presented in Table 2.

The structural features of both the general network and individual neuron types show marked increases, reflecting the natural topological richness introduced by higher connection density. Beyond structural variations, only the overall CV and the RS firing rate exhibit notable trends, implying that these two measures are relatively sensitive to changes in density. The accompanying Figure 2, based on the collected data, illustrates a consistent increase in RS neuron firing rates and a decrease in overall CV with increasing density. This suggests that in this model, when increasing the density, RS neurons fire more frequently and the model shifts towards more regular spiking activity.

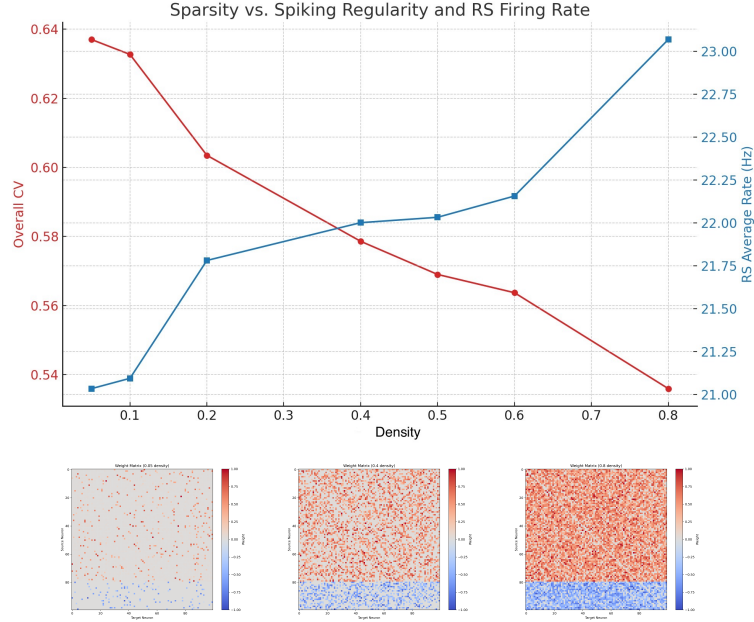


Figure 2: Effect of sparsity on overall CV and RS firing rate weight matrices with densities 0.05, 0.4, and 0.8

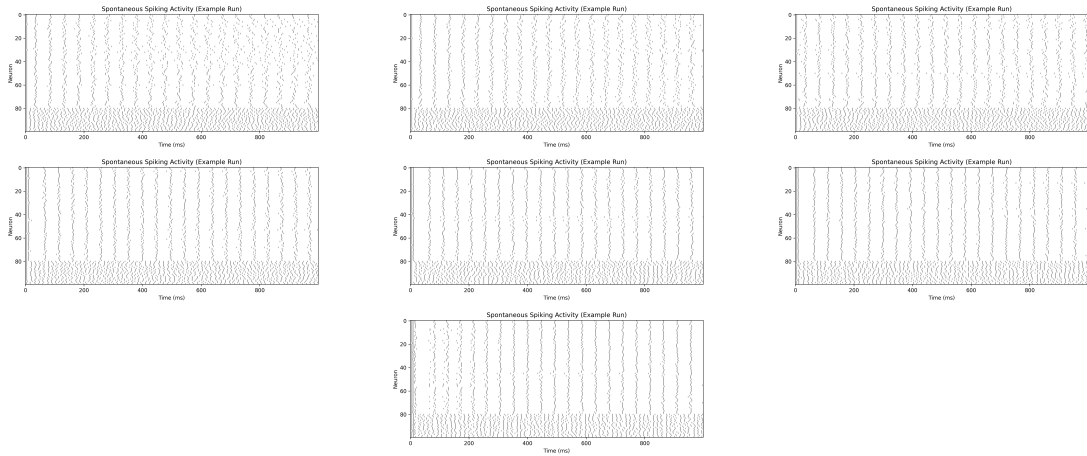


Figure 3: Spontaneous spiking activity spectrums under varying network densities densities shown: 0.05–0.8, arranged by row (top to bottom): {0.05, 0.1, 0.2}, {0.4, 0.5, 0.6}, and {0.8}.

Spontaneous spiking activity spectrums (shown in Figure 3) demonstrate that spike patterns vary with network sparsity. Using simulation by the model, FS neurons fire at high rates even under sparse conditions and show limited responsiveness to increased density. Therefore, the focus here is placed on RS neuron spiking behaviour. At the lowest density, spiking activity is largely disorganised, with neurons firing in a more isolated and asynchronous manner. There is a lack of a stable global rhythm. Although there are occasional bursts of synchronized spikes that appear transiently, they are not consistent throughout the full duration. As density increases, the network begins to exhibit clearer rhythmic spiking, and most neurons show vertically aligned spike bands, indicative of synchronised firing times. At a density of 80%, synchronisation occurs later in the simulation, and a subtle disorganisation is present during the initial 100–200 ms, which contrasts with the lower-density conditions.

3.2 Effect of Connection Strength

3.2.1 Settings

To figure out how the connection strength (scaling of different neurons) influences the spiking pattern, I completed a series of experiments. In these experiments, the network configuration was kept consistent, and the sparsity was constant (10%). To isolate the effect of excitatory and inhibitory synaptic strength, I set the excitatory scaling to $\{4, 6, 8\}$ and the inhibitory to $\{-4, -6, -8\}$, resulting in 9 pairs of combinations. Meanwhile, to test the robustness of the network's response, I further modified the noise standard deviations of the input current to $\{0.5, 1.0, 2.0\}$. Then the model was simulated exactly the same as the first.

3.2.2 Result and Analysis

To quantify the responses, I collected the average firing rate and CV for each experiment. Detailed data are presented in Table 3. These results were visualised using line plots and heat maps, as shown in Figure 4 and Figure 5.

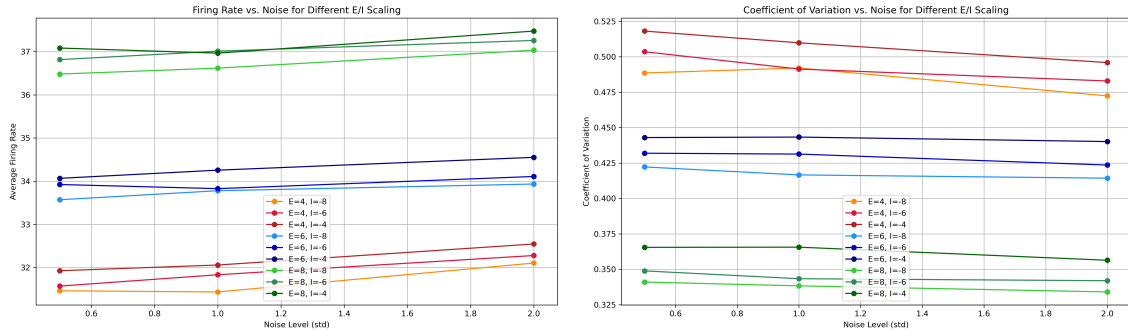


Figure 4: Line graphs of firing rate and CV vs. noise for different E/I scaling

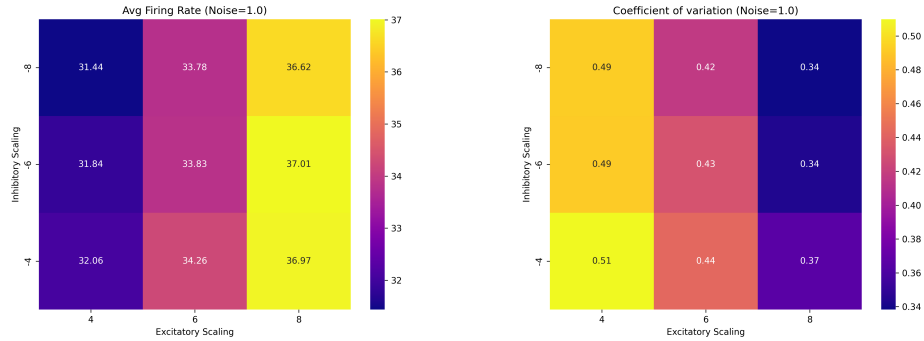


Figure 5: Heat maps of firing rate or CV across (exc, inh) scaling

The line graph on the left illustrates a general increase in firing rate with rising noise level across all excitatory/ inhibitory scaling combinations. In this model, when inhibitory scaling is kept constant, increasing excitatory scaling leads to higher firing rates, whereas increased inhibitory scaling has the opposite effect. As shown in the heat map, the firing rate increases from the bottom left to the top right, indicating that excitation predominantly drives the response but is modulated by inhibition.

In the line plot on the right, the CV tends to decrease slightly as noise level increases, indicating relatively stable spiking variability for each excitatory/ inhibitory scaling pair under perturbation. In this experiment, high levels of both excitation and inhibition tend to produce lower CVs, though excitatory scaling exerts a more pronounced effect. Strong excitation combined with strong inhibition results in the lowest CV values,

suggesting the most stable firing patterns, while the highest CV is at (E=4, I=-4), indicating more irregular spiking.

3.3 Effect of Cell Type Composition

3.3.1 Settings

To explore how different types of neural cells and their ratios affect network spiking activity, I conducted six experiments using the same model. Similarly, the overall network configuration was maintained constant throughout. The experiments varied the composition of neuron types (regular spiking (RS), fast spiking (FS), and low-threshold spiking (LTS)) to form six distinct groups: A1: {80% RS, 20% FS, 0% LTS}, A2: {60% RS, 20% FS, 20% LTS}, A3: {40% RS, 30% FS, 30% LTS}, B1: {70% RS, 30% FS, 0% LTS}, B2: {70% RS, 15% FS, 15% LTS}, and B3: {70% RS, 0% FS, 30% LTS}. Then the model was simulated the same as before.

3.3.2 Result and Analysis

In this experiment, after calculating the average firing rate and CV for different cell types, it was observed that variations in cell-type ratios had no significant effect on their average firing rates. The corresponding data are presented in Table 4. As shown in Figure 6, the average firing rate remains stable across the various ratios. Although CV varies with changes in cell-type ratios, it does not exhibit any discernible patterns. Similarly, the spontaneous spiking activity spectrums in Figure 7 show no substantial changes for each cell type. To further investigate this, Fourier analysis was conducted and Fourier power spectrums for different cell types were computed, which are illustrated in Figures 8 to Figure 13.

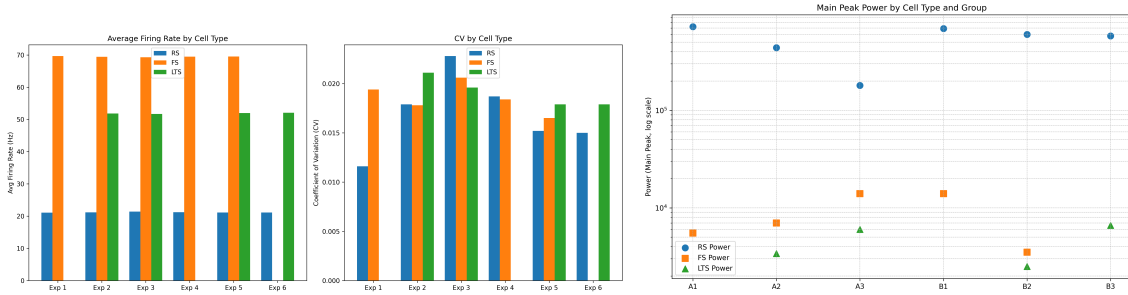


Figure 6: Comparison of firing rate, CV, and main peak power across cell-type ratios

Across all six groups, the RS population consistently exhibited strong rhythmic activity, with dominant peaks around 20 Hz and a secondary peak near 40 Hz. However, notable changes were observed in the power range of RS neurons. The graph of the main peak powers in Figure 6 conducts the power at the dominant peak for each type of cell. By comparing experimental groups A1 and B1, where the FS proportion increases from 20% to 30% while the LTS proportion remains at 0%, an increase in FS power is evident, accompanied by a slight reduction in RS power. And in groups A1 to A3, where the overall inhibitory input increases, a marked reduction in RS rhythmic power is observed. Furthermore, when maintaining the RS ratio (groups B1 to B3), the RS power shows a modest decline, while FS power decreases substantially with the introduction of LTS cells.

3.4 Effect of Weight Distribution and Topologies

3.4.1 Settings

To explore how the shape of synaptic weight distributions and network topologies influence neuronal spiking dynamics, I conducted five experiments using the same neuron model and parameters as in the original Izhikevich model. In contrast to previous experiments, each network in this series was constructed with a distinct connectivity pattern, corresponding to a specific synaptic weight matrix. The network types

examined were: a random network with normally distributed weights, a random network with log-normally distributed weights, a lattice network, a small-world network, and a randomly wired scale-free network. I used L2 normalization, so synaptic weights drawn from the specified distribution were normalized to have same mean and standard deviation to ensure fairness in comparison. The model was then simulated under identical conditions.

3.4.2 Result and Analysis

In this experiment, structural information and firing rate data were collected, and average firing rate distributions were plotted for each neuronal type. The corresponding data are presented in Table 5 and Table 6. Among the networks, the random network with a normally distributed degree exhibits the highest overall connectivity, whereas the small-world and lattice networks display the lowest degrees. Furthermore, the scale-free network shows the highest clustering coefficient, in contrast to the small-world and lattice networks, which exhibit the lowest. The mean synaptic strength is also greatest in the normally distributed random network, while it is comparatively weaker in the lattice and small-world topologies.

Despite these structural differences, the model demonstrates that both the firing rates and CV remain stable across all network types, for both the overall population and specific cell types. For all networks, the firing rate distributions of specific cell types are shown in Figure 14 to Figure 18, revealing that the average firing rate distribution of RS neurons is insensitive to changes in network connectivity. Although FS neurons exhibit distinct distributional patterns, their firing rates vary within a similar overall range.

4 Discussion

This study investigated the influence of network sparsity, synaptic weight scaling, neuronal composition, and variations in weight distributions and topologies on population-level spiking behaviour in Izhikevich model networks. The following sections present a detailed discussion of the findings in relation to each research question.

- Connection density:** Within the scope of this experiment, increased network density was found to enhance overall synchrony, which is consistent with the findings of Kim and Lim [2014] and Harris et al. [2023], who report that reduced connection density is associated with sparse synchronisation. In this study, only RS neurons exhibited firing rate changes in response to varying sparsity, suggesting that RS cells may be more sensitive to network density than FS cells. Furthermore, at the density of 80%, the spiking raster initially appeared disordered, but gradually transitioned toward synchrony, potentially indicating a dynamic, adaptive settling process in densely connected networks. As this study tested densities up to 80% only, further investigation into values slightly above and below this threshold would be valuable in assessing this hypothesis more thoroughly.
- Connection strength:** Under the constraints of this experiment, the firing rate generally increased with excitation, decreased with inhibition; CV dropped when both were strong, aligning with the result of Sadeh and Clopath [2021], which states that synaptic strength of inhibitory connections affect both the firing rates and the stability of spiking assemblies. Although firing rate and CV across different excitatory/ inhibitory ratio groups remained largely stable under varying noise, some notable reversals were observed. When $E = 8$ and noise level was at 1.0, the firing rate reversed at $I = -6$ and -4 ; with $E = 4$, the CV reversed at $I = -6$ and -8 when noise levels were at 1.0 and 2.0. These fluctuations may support earlier findings by Rubin et al. [2017], where imbalances in excitatory/ inhibitory strength are associated with noise-sensitive dynamics. However, further investigation is necessary to determine whether these observed reversals indeed result from the imbalance. It is possible that the parameter ranges employed in our model did not reach the thresholds required to elicit clear signs of instability. Therefore, incorporating stronger or more structured noise inputs may be necessary to uncover the sensitivities proposed in previous studies.
- Cell type composition:** In experiment investigating the influence of cell-type composition on network dynamics, power spectrum analysis revealed more nuanced effects on rhythmic structure. In networks composed only of FS and RS neurons, a greater proportion of FS cells reduced the power of RS firing

rates, suggesting a suppression effect driven by FS neurons. Furthermore, when the RS population was restricted while modifying the ratios of LTS and FS neurons, power fluctuations were observed in both inhibitory neuron types, which indicates potential interactions between them. This result may extend the research of Tomov et al. [2014], which investigated how changing ratios of excitatory types could affect firing rate with certain inhibitory subtype. Notably, the inclusion of LTS neurons attenuated RS oscillations without eliminating their dominant frequency components, which supports the claim that the introduction of LTS reduces the probability of RS spike at higher drive frequencies from Vierling-Claassen et al. [2010]. However, our results contrast with the conclusions of Destexhe [2009] which suggest that LTS neurons promote asynchronous firing. The result shows that the overall CV decreases with the higher ratios of LTS, indicating more regular firing. These discrepancies may arise from specific model configurations or parameter choices, such as network size or the strength of recurrent excitation.

- **Weight distribution and topologies** In this experiment, normalization was applied to ensure comparability across networks with different structures. I chose L2 normalization over L1 because L1 normalization can reduce values close to zero to exact zeros, potentially discarding important information. However, L2 normalization may not be the optimal choice either. In particular, the minimal differences observed between the lattice and small-world networks suggest that L2 normalization may have obscured subtle structural distinctions, making cross-network comparisons less precise. Developing a more appropriate normalization or comparison metric could improve interpretability in future analyses. In the last experiment, network topology and synaptic weight distribution had negligible effects on RS spiking, and only limited influence on FS activity. These results are consistent with the findings of Kriener et al. [2009], who reported that network topology and weight distribution had little impact on firing rates. Due to time constraints, the current experiment was limited in scope. Future work could explore networks combining different features, such as scale-free weight distributions together with small-world connectivity, which may offer deeper insights into their joint influence on network dynamics and more closely resemble the structure of real neural networks.

A Additional Tables and Figures

Table 2: Data from connection density

(a) Overall network metrics

Sparsity	Avg Rate	CV	Avg Degree	Clustering	Strength Mean	Strength SEM
0.05	30.859	0.637	6.495	0.0208	3.247	0.129
0.10	30.848	0.633	11.775	0.0322	5.887	0.175
0.20	31.184	0.603	21.475	0.0626	10.737	0.226
0.40	30.951	0.579	42.121	0.1320	21.060	0.283
0.50	30.787	0.569	52.423	0.1746	26.212	0.288
0.60	30.845	0.564	62.323	0.2137	31.161	0.292
0.80	31.502	0.536	81.824	0.3057	40.912	0.315

(b) RS and FS cell-specific metrics

Sparsity	RS Rate	RS CV	RS Strength	RS SEM	FS Rate	FS CV	FS Strength	FS SEM
0.05	21.035	0.0083	3.465	0.140	70.155	0.0181	2.379	0.241
0.10	21.095	0.0136	6.159	0.187	69.860	0.0189	4.803	0.355
0.20	21.781	0.0185	11.111	0.237	68.795	0.0195	9.243	0.484
0.40	22.001	0.0005	21.612	0.290	66.750	0.0184	18.854	0.594
0.50	22.033	0.0072	26.719	0.304	65.805	0.0181	24.183	0.582
0.60	22.156	0.0152	31.841	0.283	65.600	0.0175	28.444	0.617
0.80	23.068	0.0200	41.750	0.299	65.240	0.0174	37.561	0.565

Table 3: Data from connection strength

exc	inh	noise	avg_rate	cv	exc	inh	noise	avg_rate	cv
4	-4	0.5	31.928	0.518	6	-6	0.5	33.924	0.432
4	-4	1.0	32.060	0.510	6	-6	1.0	33.828	0.431
4	-4	2.0	32.547	0.496	6	-6	2.0	34.110	0.424
6	-4	0.5	34.067	0.443	8	-6	0.5	36.815	0.349
6	-4	1.0	34.256	0.443	8	-6	1.0	37.011	0.343
6	-4	2.0	34.553	0.440	8	-6	2.0	37.259	0.342
8	-4	0.5	37.083	0.365	4	-8	0.5	31.463	0.489
8	-4	1.0	36.965	0.366	4	-8	1.0	31.437	0.492
8	-4	2.0	37.476	0.356	4	-8	2.0	32.106	0.472
4	-6	0.5	31.572	0.504	6	-8	0.5	33.571	0.422
4	-6	1.0	31.835	0.491	6	-8	1.0	33.780	0.417
4	-6	2.0	32.280	0.483	6	-8	2.0	33.935	0.414
					8	-8	0.5	36.481	0.341
					8	-8	1.0	36.618	0.338
					8	-8	2.0	37.034	0.334

Table 4: Data from cell type composition

Metric	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	Exp 6
overall_avg_rate	30.790	36.959	44.866	35.682	33.009	30.407
overall_cv	0.632	0.544	0.454	0.620	0.569	0.467
overall_avg_degree	11.740	11.246	10.616	11.452	11.757	11.681
overall_clustering	0.0319	0.0355	0.0424	0.0330	0.0352	0.0348
overall_strength_mean	5.870	5.623	5.308	5.726	5.879	5.841
overall_strength_sem	0.167	0.175	0.184	0.169	0.173	0.177
FS_avg_rate	69.665	69.415	69.313	69.470	69.507	—
FS_cv	0.0194	0.0178	0.0206	0.0184	0.0165	—
FS_strength_mean	4.798	4.746	4.704	4.768	4.669	—
FS_strength_sem	0.366	0.339	0.292	0.280	0.392	—
LTS_avg_rate	—	51.820	51.690	—	51.967	52.090
LTS_cv	—	0.0211	0.0196	—	0.0179	0.0179
LTS_strength_mean	—	4.829	4.511	—	4.923	4.844
LTS_strength_sem	—	0.360	0.274	—	0.411	0.286
RS_avg_rate	21.071	21.187	21.413	21.201	21.126	21.114
RS_cv	0.0116	0.0179	0.0228	0.0187	0.0152	0.0150
RS_strength_mean	6.138	6.180	6.358	6.136	6.342	6.268
RS_strength_sem	0.175	0.209	0.272	0.188	0.190	0.201

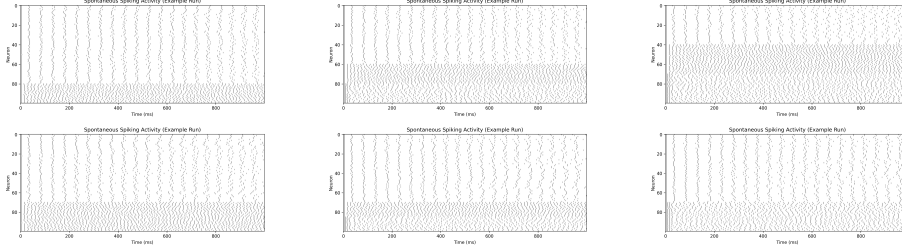


Figure 7: Spontaneous spiking activity spectrum under varying cell type compositions

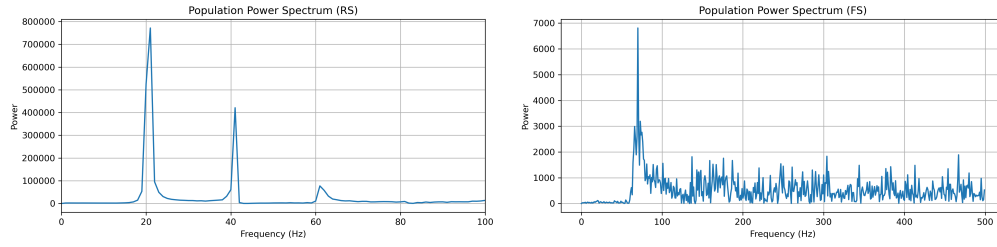


Figure 8: Fourier power spectrum of A1

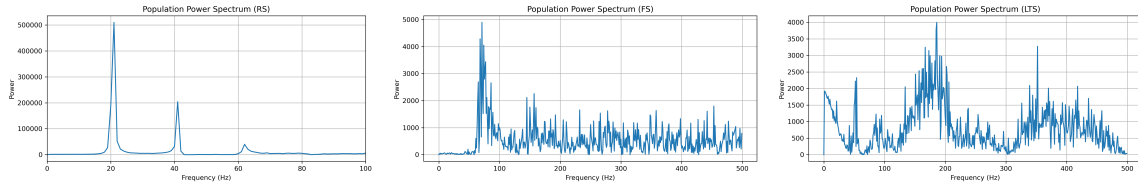


Figure 9: Fourier power spectrum of A2

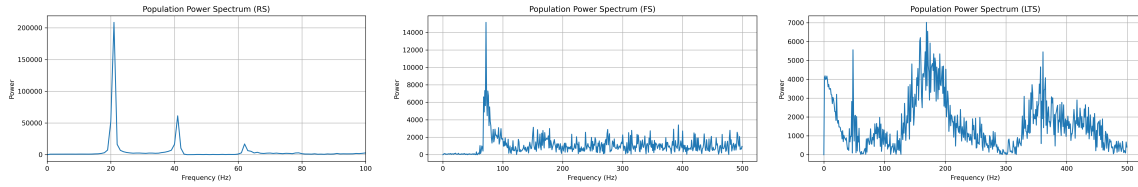


Figure 10: Fourier power spectrum of A3

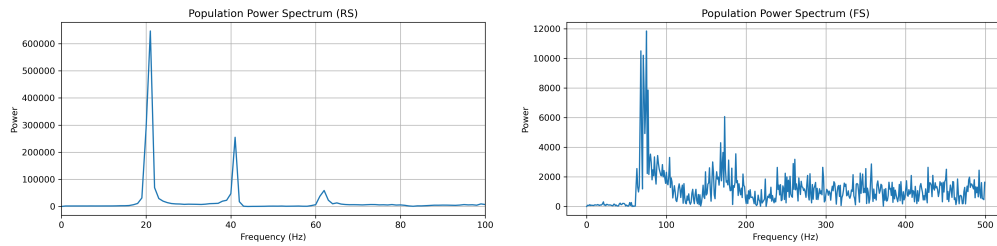


Figure 11: Fourier power spectrum of B1

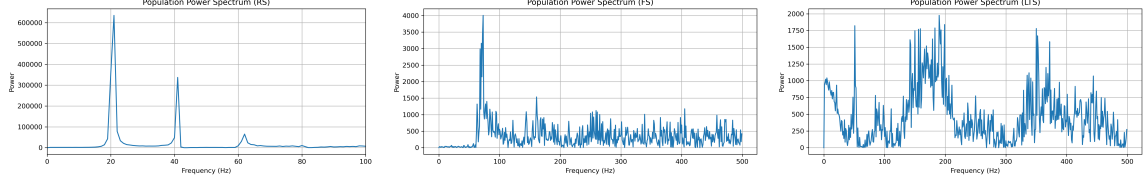


Figure 12: Fourier power spectrum of B2

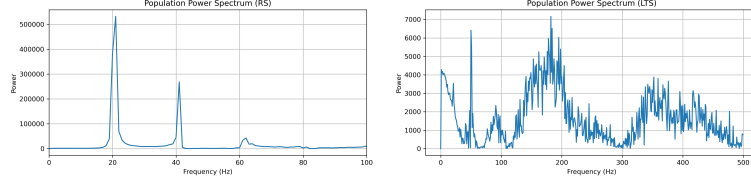


Figure 13: Fourier power spectrum of B3

Table 5: Overall network metrics for weight distribution and topologies

Distribution	Rate	CV	Degree	Clustering	Strength Mean
Normal	30.869	0.6378	7.5813	0.0410	3.7907
Log-normal	30.885	0.6388	6.2754	0.0247	3.1377
Lattice	30.794	0.6355	5.2255	0.0181	2.6127
Small-world	30.814	0.6359	5.2386	0.0177	2.6193
Scale-free	30.866	0.6380	6.1470	0.0476	3.0735

Table 6: Cell-specific metrics for weight distribution and topologies

Dist.	FS Rate	FS CV	FS Str.	RS Rate	RS CV	RS Str.
Normal	70.225	0.0190	2.7105	21.0300	0.0076	4.0607
Log-normal	70.330	0.0171	2.1411	21.0238	0.0062	3.3868
Lattice	69.920	0.0169	1.5639	21.0125	0.0050	2.8750
Small-world	69.990	0.0161	1.5865	21.0200	0.0065	2.8775
Scale-free	70.240	0.0159	2.0088	21.0225	0.0067	3.3397

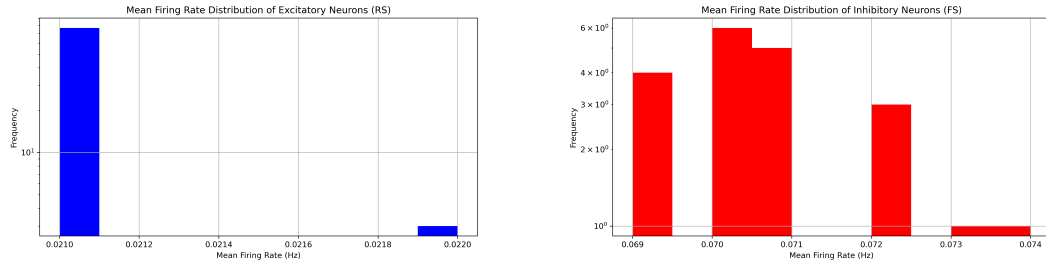


Figure 14: Mean firing rate distribution spectrum of random network with normal distribution

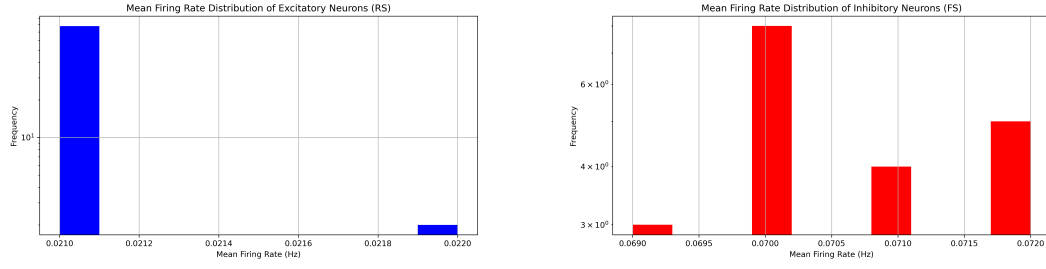


Figure 15: Mean firing rate distribution spectrum of random network with log-normal distribution

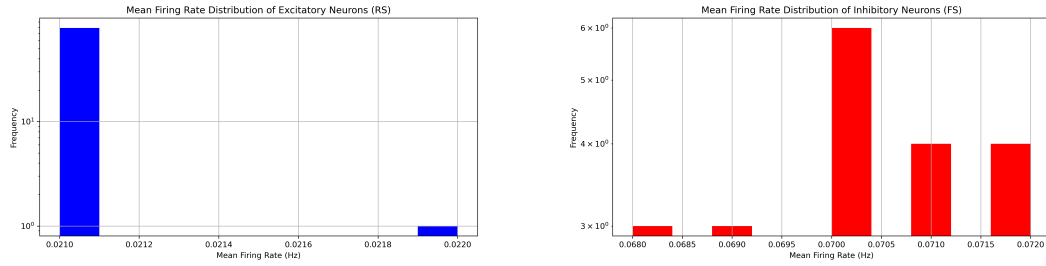


Figure 16: Mean firing rate distribution spectrum of lattice network

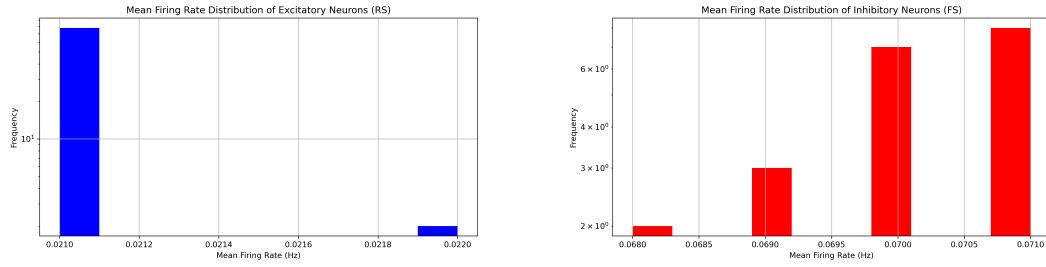


Figure 17: Mean firing rate distribution spectrum of small-world network

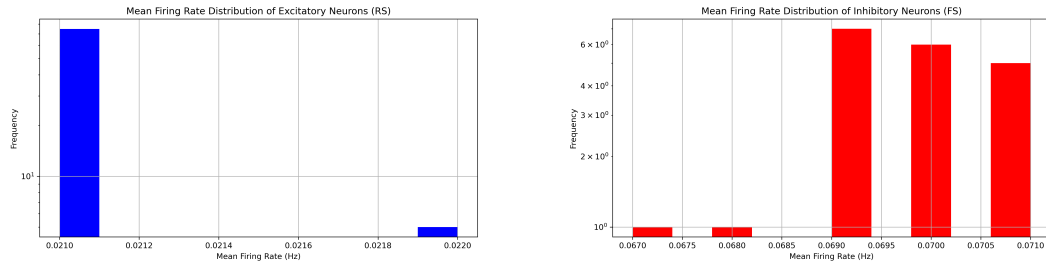


Figure 18: Mean firing rate distribution spectrum of randomly wired scale-free network

References

Ed Bullmore and Olaf Sporns. Complex brain networks: graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience*, 10(3):186–198, March 2009. ISSN 1471-003X, 1471-0048. doi:

- 10.1038/nrn2575. URL <https://www.nature.com/articles/nrn2575>. Publisher: Springer Science and Business Media LLC.
- Tim P. Vogels, Kanaka Rajan, and L.F. Abbott. NEURAL NETWORK DYNAMICS. *Annual Review of Neuroscience*, 28(1):357–376, July 2005. ISSN 0147-006X, 1545-4126. doi: 10.1146/annurev.neuro.28.061604.135637. URL <https://www.annualreviews.org/doi/10.1146/annurev.neuro.28.061604.135637>. Publisher: Annual Reviews.
- E.M. Izhikevich. Simple model of spiking neurons. *IEEE Transactions on Neural Networks*, 14(6):1569–1572, November 2003. ISSN 1045-9227. doi: 10.1109/tnn.2003.820440. URL <http://ieeexplore.ieee.org/document/1257420/>. Publisher: Institute of Electrical and Electronics Engineers (IEEE).
- Peter Dayan and Laurence F. Abbott. *Theoretical neuroscience: computational and mathematical modeling of neural systems*. Computational neuroscience. MIT Press, Cambridge, Mass., 2001. ISBN 978-0-262-04199-7 978-0-262-54185-5.
- Jeffrey S. Isaacson and Massimo Scanziani. How Inhibition Shapes Cortical Activity. *Neuron*, 72(2):231–243, October 2011. ISSN 0896-6273. doi: 10.1016/j.neuron.2011.09.027. URL <https://linkinghub.elsevier.com/retrieve/pii/S0896627311008798>. Publisher: Elsevier BV.
- Eleonora Catsigeras. Dale’s Principle is necessary for an optimal neuronal network’s dynamics. *Applied Mathematics*, 04(10):15–29, 2013. ISSN 2152-7385, 2152-7393. doi: 10.4236/am.2013.410A2002. URL <http://arxiv.org/abs/1307.0597>. arXiv:1307.0597 [physics].
- Barry W. Connors and Michael J. Gutnick. Intrinsic firing patterns of diverse neocortical neurons. *Trends in Neurosciences*, 13(3):99–104, March 1990. ISSN 0166-2236. doi: 10.1016/0166-2236(90)90185-d. URL <https://linkinghub.elsevier.com/retrieve/pii/016622369090185D>. Publisher: Elsevier BV.
- Jay R. Gibson, Michael Beierlein, and Barry W. Connors. Two networks of electrically coupled inhibitory neurons in neocortex. *Nature*, 402(6757):75–79, November 1999. ISSN 0028-0836, 1476-4687. doi: 10.1038/47035. URL <https://www.nature.com/articles/47035>. Publisher: Springer Science and Business Media LLC.
- Jeffrey C. Magee and Christine Grienberger. Synaptic Plasticity Forms and Functions. *Annual Review of Neuroscience*, 43(1):95–117, July 2020. ISSN 0147-006X, 1545-4126. doi: 10.1146/annurev-neuro-090919-022842. URL <https://www.annualreviews.org/doi/10.1146/annurev-neuro-090919-022842>. Publisher: Annual Reviews.
- Rieke Fruengel and Marcel Oberlaender. Sparse connectivity enables efficient information processing in cortex-like artificial neural networks. *Frontiers in Neural Circuits*, 19, March 2025. ISSN 1662-5110. doi: 10.3389/fncir.2025.1528309. URL <https://www.frontiersin.org/articles/10.3389/fncir.2025.1528309/full>. Publisher: Frontiers Media SA.
- Sen Song, Per Jesper Sjöström, Markus Reigl, Sacha Nelson, and Dmitri B Chklovskii. Highly Nonrandom Features of Synaptic Connectivity in Local Cortical Circuits. *PLoS Biology*, 3(3):e68, March 2005. ISSN 1545-7885. doi: 10.1371/journal.pbio.0030068. URL <https://dx.plos.org/10.1371/journal.pbio.0030068>. Publisher: Public Library of Science (PLOS).
- Duncan J Watts and Steven H Strogatz. Collective dynamics of ‘small-world’ networks. 1998.
- Albert-László Barabási and Réka Albert. Emergence of Scaling in Random Networks. *Science*, 286(5439):509–512, October 1999. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.286.5439.509. URL <https://www.science.org/doi/10.1126/science.286.5439.509>. Publisher: American Association for the Advancement of Science (AAAS).

- A. L. Hodgkin and A. F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500–544, August 1952. ISSN 0022-3751. doi: 10.1113/jphysiol.1952.sp004764.
- Sang-Yoon Kim and Woonchang Lim. Effect of Small-World Connectivity on Fast Sparsely Synchronized Cortical Rhythms, September 2014. URL <http://arxiv.org/abs/1403.1034>. arXiv:1403.1034 [q-bio].
- Isabelle D. Harris, Hamish Meffin, Anthony N. Burkitt, and Andre D. H. Peterson. Effect of sparsity on network stability in random neural networks obeying Dale’s law. *Physical Review Research*, 5(4), November 2023. ISSN 2643-1564. doi: 10.1103/physrevresearch.5.043132. URL <https://link.aps.org/doi/10.1103/PhysRevResearch.5.043132>. Publisher: American Physical Society (APS).
- Sadra Sadeh and Claudia Clopath. Excitatory-inhibitory balance modulates the formation and dynamics of neuronal assemblies in cortical networks. *SCIENCE ADVANCES*, 2021.
- Ran Rubin, L. F. Abbott, and Haim Sompolinsky. Balanced excitation and inhibition are required for high-capacity, noise-robust neuronal selectivity. *Proceedings of the National Academy of Sciences*, 114(44), October 2017. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.1705841114. URL <https://pnas.org/doi/full/10.1073/pnas.1705841114>. Publisher: Proceedings of the National Academy of Sciences.
- Alain Destexhe. Self-sustained asynchronous irregular states and Up/Down states in thalamic, cortical and thalamocortical networks of nonlinear integrate-and-fire neurons. *Journal of Computational Neuroscience*, 27(3):493–506, December 2009. ISSN 0929-5313, 1573-6873. doi: 10.1007/s10827-009-0164-4. URL <http://arxiv.org/abs/0809.0654>. arXiv:0809.0654 [q-bio].
- Dorea Vierling-Claassen, Jessica A. Cardin, Christopher I. Moore, and Stephanie R. Jones. Computational Modeling of Distinct Neocortical Oscillations Driven by Cell-Type Selective Optogenetic Drive: Separable Resonant Circuits Controlled by Low-Threshold Spiking and Fast-Spiking Interneurons. *Frontiers in Human Neuroscience*, 4, 2010. ISSN 1662-5161. doi: 10.3389/fnhum.2010.00198. URL <http://journal.frontiersin.org/article/10.3389/fnhum.2010.00198/abstract>. Publisher: Frontiers Media SA.
- Petar Tomov, Rodrigo F. O. Pena, Michael A. Zaks, and Antonio C. Roque. Sustained oscillations, irregular firing, and chaotic dynamics in hierarchical modular networks with mixtures of electrophysiological cell types. *Frontiers in Computational Neuroscience*, 8, September 2014. ISSN 1662-5188. doi: 10.3389/fncom.2014.00103. URL <http://journal.frontiersin.org/article/10.3389/fncom.2014.00103/abstract>. Publisher: Frontiers Media SA.
- H. Y. Li, G. M. Cheng, and Emily S. C. Ching. Heterogeneous Responses to Changes in Inhibitory Synaptic Strength in Networks of Spiking Neurons. *Frontiers in Cellular Neuroscience*, 16, February 2022. ISSN 1662-5102. doi: 10.3389/fncel.2022.785207. URL <https://www.frontiersin.org/articles/10.3389/fncel.2022.785207/full>. Publisher: Frontiers Media SA.
- Murray Shanahan. On the Dynamical Complexity of Small-World Networks of Spiking Neurons.
- Birgit Kriener, Moritz Helias, Ad Aertsen, and Stefan Rotter. Correlations in spiking neuronal networks with distance dependent connections. *Journal of Computational Neuroscience*, 27(2): 177–200, October 2009. ISSN 0929-5313, 1573-6873. doi: 10.1007/s10827-008-0135-1. URL <http://link.springer.com/10.1007/s10827-008-0135-1>. Publisher: Springer Science and Business Media LLC.
- Henry Markram, Maria Toledo-Rodriguez, Yun Wang, Anirudh Gupta, Gilad Silberberg, and Caizhi Wu. Interneurons of the neocortical inhibitory system. *Nature Reviews Neuroscience*, 5(10):793–807, October 2004. ISSN 1471-003X, 1471-0048. doi: 10.1038/nrn1519. URL <https://www.nature.com/articles/nrn1519>. Publisher: Springer Science and Business Media LLC.

Robert Tibshirani. Regression Shrinkage and Selection Via the Lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288, January 1996. ISSN 1369-7412, 1467-9868. doi: 10.1111/j.2517-6161.1996.tb02080.x. URL <https://academic.oup.com/jrsssb/article/58/1/267/7027929>. Publisher: Oxford University Press (OUP).

Benoît Siri, Mathias Quoy, Bruno Delord, Bruno Cessac, and Hugues Berry. Effects of Hebbian learning on the dynamics and structure of random networks with inhibitory and excitatory neurons. *Journal of Physiology-Paris*, 101(1-3):136–148, January 2007. ISSN 0928-4257. doi: 10.1016/j.jphysparis.2007.10.003. URL <https://linkinghub.elsevier.com/retrieve/pii/S0928425707000356>. Publisher: Elsevier BV.